

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO3: *Analyze shape representation methods and techniques such as contours, regions, deformable models, and multi-resolution analysis. (BL4)*

Assessment Tool Weightage: 30%

QUESTIONS: 2

Total Marks:30

Annexure A
Case study

Code of Conduct:

I Peram.Mahitha(192524136)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:P.Mahitha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Annexure A

Case study

Case Study 1: Defect Detection in Industrial Parts

A manufacturing company wants to automatically detect defects in gears and machine components using computer vision. The images of the components are binary and may contain small protrusions or missing parts.

Questions:

- A. How can region-based representation and medial axis computation help in identifying defective parts?
- B. Discuss how noise in image acquisition could affect shape analysis and propose preprocessing steps to improve accuracy.
- C. Compare the effectiveness of Fourier descriptors versus wavelet descriptors for classifying normal vs. defective components.

Ans) Defect Detection in Industrial Parts

A. Region-Based Representation and Medial Axis

- **Region-based representation** considers the complete area of the gear/component rather than only its boundary.
- Features such as **area, perimeter, centroid, circularity, and shape moments** can be extracted.
- A **medial axis (skeleton)** reduces the object to its central structure while preserving its overall shape.
- For a normal gear, the skeleton has a consistent structure and expected number of branches/teeth.
- A **missing tooth or protrusion** changes the skeleton, branch points, length, or local thickness.
- By comparing the extracted features with those of a normal component, the system can identify **defective parts automatically**.

Example:

A missing gear tooth produces a change in the boundary and also creates a difference in the skeleton structure. Hence, medial-axis analysis can detect the defect.

B. Effect of Noise and Preprocessing

Noise during image acquisition can create false protrusions, holes, broken boundaries, or unwanted small regions. This can lead to incorrect area, perimeter, skeleton, and shape measurements.

Suitable preprocessing steps:

1. Convert to grayscale if the input is not already binary.
2. Apply thresholding to obtain a clean binary image.
3. Use median filtering to remove salt-and-pepper noise.
4. Apply morphological opening to remove small unwanted objects/protrusions.
5. Apply morphological closing to fill small gaps and holes.
6. Remove small connected components that are not part of the actual object.

7. Perform skeletonization only after cleaning the binary image.

This produces a cleaner shape and improves the reliability of defect detection.

C. Fourier Descriptors vs. Wavelet Descriptors

Feature	Fourier Descriptors	Wavelet Descriptors
Representation	Mainly represents overall contour	Represents contour at multiple scales
Global shape	Excellent	Good
Local defects	Less effective	More effective
Small protrusions	May be difficult to detect	Can detect well
Missing parts	Good for major shape changes	Good for local changes
Noise sensitivity	Can be reduced using selected coefficients	Can analyze different frequency components
Complexity	Relatively simple	More complex

Fourier descriptors are useful for describing the overall shape of a normal gear and identifying large deviations from it.

Wavelet descriptors are more suitable when defects are small and localized, such as tiny cracks, protrusions, or missing portions, because wavelets provide both spatial and frequency information.

Conclusion

For industrial gear defect detection, region-based representation and medial-axis analysis provide useful structural information. Proper preprocessing is essential to prevent noise from being interpreted as defects. Between the two descriptors, Fourier descriptors are better for global shape classification, while wavelet descriptors are generally more effective for detecting small local defects. A combination of both can provide more reliable classification.

Case Study 2: Medical Image Segmentation

A hospital is developing a system to segment organs from CT scans. The organ boundaries are irregular and may vary slightly between patients.

Questions:

- A. Explain how active contours (snakes) can be applied to segment the organ boundaries.
- B. Discuss the challenges of initializing contours and tuning parameters for accurate segmentation.
- C. How could level set methods improve segmentation for cases where organs split or merge in the images?

Ans)

A. Active Contours (Snakes) for Organ Segmentation

- **Active contours**, also called **snakes**, are curves that move toward the actual boundary of an organ.
- An initial contour is placed **inside or around the organ**.
- The contour is then iteratively adjusted based on image information.
- The snake uses two main forces:
 - **Internal force**: keeps the contour smooth and prevents unrealistic shapes.
 - **External force**: attracts the contour toward strong edges or intensity changes.
- The process continues until the contour reaches the **organ boundary**.
- This is useful for CT images because organ boundaries can be **irregular and slightly different between patients**.

Example:

For liver segmentation, an initial contour is placed around the liver. The snake gradually moves toward the liver boundary and produces the final segmented region.

B. Challenges in Initialization and Parameter Tuning

1. Initialization challenges

- The initial contour must be reasonably close to the organ.
- If it is placed too far away, the snake may converge to the **wrong boundary**.
- Multiple nearby edges, such as blood vessels or bones, can confuse the contour.
- Different patients may have different organ shapes and sizes, making a fixed initialization difficult.

2. Parameter tuning challenges

- **Smoothness parameter**: controls how smooth the contour should be.
- **Edge attraction parameter**: determines how strongly the contour follows image edges.
- **Balloon force**: controls whether the contour expands or contracts.
- If parameters are too strong, the contour may **over-smooth** the organ.
- If parameters are too weak, the contour may become irregular or stop before reaching the boundary.

Solution:

Use automatic initialization, image preprocessing, multi-scale processing, and patient-specific parameter selection to improve segmentation accuracy.

C.Level Set Methods for Splitting or Merging Organs

- A **level set method** represents the contour as a mathematical surface rather than as a fixed curve.
- Unlike traditional snakes, the contour can **change its topology automatically**.
- This means one contour can **split into multiple regions**, or multiple contours can **merge into one region**.
- This is especially useful when organs or anatomical structures appear connected or separated in different CT slices.
- Level sets can also handle **irregular boundaries** and complex shapes more naturally.
- They are therefore more flexible than traditional active contours for complicated medical images.

Example:

If two connected anatomical regions need to be separated, a level-set contour can automatically split into two regions. Similarly, separate regions can merge when the image structure indicates that they belong together.

Conclusion

Active contours are effective for following **irregular organ boundaries**, but their performance depends heavily on initialization and parameter selection. **Level set methods provide greater flexibility** because they can automatically handle splitting and merging of regions, making them suitable for complex CT organ segmentation.